Probability - 1

#Types of Events :

1) **Equally Likely** ⇒ When all outcomes have equal chances of occurrence.

2) **Mutually Exclusive / Disjoint / Incompatible** ⇒ Events whose simultaneous occurrence is impossible.

i.e. $\boxed{P(E_1) \cap P(E_2) = \phi}$

3) **Mutually Exhaustive** ⇒ Events whose union gives us sample space only.

i.e. $\boxed{P(E_1) \cup P(E_2) = S}$

#Classical Probability :

$$\boxed{P(E) = \frac{\text{No. of favourable Outcomes}}{\text{No. of Total Outcomes}}}$$

↳ We can apply this formula only when
(i) All events are equally likely
(ii) All events are mutually exclusive.

Important NOTE for entire probability :→

1) Sum of all sample points = 1

2) $\boxed{\text{Odds in favour of } E = \frac{\text{favourable outcome}}{\text{unfavourable outcome}}}$

3) $\boxed{\text{Odds against } E = \frac{\text{Unfavourable outcomes}}{\text{Favourable outcomes}}}$

4) Probability of non-occurrence of event $E = \boxed{P(\bar{E}) = P(E^c) = 1 - P(E)}$

NOTE : 1) Jab tak process complete na ho, tab tak prob. Ko multiply karte hai.

2) When classical formula not applicable, then find prob. of each sample point & add the prob. of favourable ones.

3) Ques mein '**ATLEAST**' dikhe to :
Atleast one = Total − (Exactly zero)
Atleast Two = Total − (Exac. One + Exac. zero)

Ques Two diff. Numbers taken from {0, 1 ... 10} (Total = 11) then prob. that their sum and absolute difference, both are multiple of 4 is ____.

Sol'n Pairs for which sum & diff are multiple of 4 ⇒ (0,4) (0,8) (4,8) (2,6) (2,10) (6,10)

$\text{Total} = {}^{11}C_2 = \frac{11 \cdot 10}{2} \Rightarrow 55$; $\boxed{P(E) = \frac{6}{55}}$ Ans

Ques Unbiased dice rolled. If Heads, then pair of dice rolled & Sum of no. noted. If Tails, then card from a deck of (1-9) numbered cards picked, & no. of card noted. Find prob. that noted no. is either 7 or 8.

Sol'n
Coin → Head (P = 1/2) → Pair of dice, □ + □ = ____ (7 or 8)
Coin → Tails (P = 1/2) → {1, 2, ... 9}, one card picked (7 or 8) (P = 2/9)

(*) Prob. that sum of dices is 7 ⇒
(1,6) (6,1)
(2,5) (5,2)
(4,3) (3,4)
⇒ $P(7) = \frac{6}{36}$

(*) Prob. that sum of dices is 8 ⇒
(2,6) (6,2)
(3,5) (5,3)
(4,4)
⇒ $P(8) = \frac{5}{36}$

So,
Total prob. of (7 or 8) $= \left(\frac{1}{2}\right)\left(\frac{6}{36}\right) + \left(\frac{1}{2}\right)\left(\frac{5}{36}\right) + \left(\frac{1}{2}\right)\left(\frac{2}{9}\right)$
$= \frac{19}{72}$ Ans

Ques For dices D_1, D_2, D_3 & D_4 rolled together. Prob. that D_4 shows a no. appearing on one of D_1, D_2 & D_3 is ____.

Sol'n D_1 D_2 D_3 | D_4 ; Total = 6^4

Case-01 ⇒ All three diff

$$P = {}^6C_3 \times (3!) \times 3$$

(${}^6C_3$ ↳ Selection of no. for first 3 dices; $3!$ ↳ Arrangement of no. for D_1, D_2, D_3; 3 ↳ No. of ways for D_4)

Case-02 ⇒ 1 diff, 2 alike

$$P = {}^6C_2 \times {}^2C_1 \times \left(\frac{3!}{2!}\right) \times 2$$

(${}^6C_2$ ↳ Selection of unique 2 no. for 1 diff, 2 alike; ${}^2C_1$ ↳ From 2 unique, selection of one no. that we want alike; $\frac{3!}{2!}$ ↳ Arrangement; 2 ↳ No. of ways for D_4)

Case-03 ⇒ All alike [P = 6] $\left({}^6C_1 \times 1 \times 1\right)$

$$P(E) = \frac{P(\text{case1}) + P(\text{case2}) + P(\text{case3})}{6^4}$$

$\boxed{P(E) = \frac{91}{216}}$ Ans

Ques (JEE Adv 2014) 3 boys & 2 girls stand in a queue. Prob. that no. of boys ahead of every girl is atleast one more than no. of girls ahead her is ____.

Sol'n No. of the Boys KPSI ladki ke aage khade hai = (Kis ki ladki ke aage fifth ladkiya hai) + 1 + 2 + 3

In Bracket: (Aage Kitni ladkiya hai , Aage Kitne ladke hai)

(2,3) (0,3) ← In Bracket
G G B B B
(1,2) (0,2)
B G G B B
(1,3) (0,2)
G B G B B

(0,3)
G B B G B
(1,2) (0,1)
B G B G B

Five possible combinations

Ques There are 6 vertices of a regular hexagon. Prob. that △ formed by joining vertices is an equilateral △ is ____.

Sol'n 1, 2, 3, 4, 5, 6

Only △ formed by joining odd/even numbered vertices would be equilateral.

So, Eq. △ = 2 ; Total △ = ${}^6C_3 = \frac{6 \cdot 5 \cdot 4}{2 \cdot 3} = 20$

$\Rightarrow \boxed{P(E) = \frac{2}{20} = \frac{1}{10}}$ Ans

Ques Fair die is rolled until two 4s obtained in succession. Prob. that experiment ends in fifth throw is ____.

Sol'n _ _ _ 4 4 ; ↳ 4 can never come in 3rd throw.

Case 1 : When no '4' obtained
$5 \times 5 \times 5 = 125$

Case 2 : When one '4' obtained
${}^2C_1 \times 1 \times 5 \times 5 \Rightarrow 50$
↳ Selection of position for '4'.

Total $P(E) = \frac{125 + 50}{6^5}$

$\boxed{P(E) = \frac{175}{6^5}}$ Ans

Question on Infinite Trials :

→ Consider n-trials of given event
→ Derive inequality acc. to ques.
→ Find value of 'n' by Hit & Total method

1) Prob. of a man hitting a target is 1/10. Least no. of shots req. so that prob. of his hitting the target at least once is greater than 1/4, is ____.

Sol'n Let the man make n shots

n times

Prob. of hitting at least once = 1 − P(Exactly zero)
$= 1 - \left(\frac{9}{10}\right)^n$

Acc. to Ques

$$1 - \left(\frac{9}{10}\right)^n > \frac{1}{4} \Rightarrow 1 - \frac{1}{4} > \left(\frac{9}{10}\right)^n$$

$$\frac{3}{4} > \frac{9^n}{10^n} \Rightarrow 3(10^n) > 4(9^n)$$

Means, the values satisfying inequality would be $n = 3, 4, 5 \cdots \infty$

$n = 1 \Rightarrow 30 > 36$ ✗
$n = 2 \Rightarrow 300 > 324$ ✗
$n = 3 \Rightarrow 3000 > 2916$ ✓

$\boxed{n_{\text{Least}} = 3}$ Ans

2) Min. no. of times a fair coin needs to be tossed, so that prob. of getting 2 heads is

atleast 0.96 ps ____.

Sol^n Let n tosses of coins: ---- (1, 2, 3, ---- n)

Prob. of atleast 2 heads = 1 - (Exactly zero head) - (Exactly 1 head)

$= 1 - \left(\frac{1}{2}\right)^n - {}^nC_1 \left(\frac{1}{2}\right)\left(\frac{1}{2}\right)^{n-1}$

(Selection of 1 from n tosses on which we want our 1 head)

$P(\text{Atleast 2H}) = 1 - \left(\frac{1}{2}\right)^n - n\left(\frac{1}{2}\right)^n$

$= 1 - \left(\frac{1+n}{2^n}\right)$

Acc. to Ques

$1 - \left(\frac{1+n}{2^n}\right) \geq 0.96 \Rightarrow \left(\frac{1+n}{2^n}\right) \leq 0.04$

$\Rightarrow 2^n \cdot (4) \geq (1+n)100$

$2^{2+n} \geq (1+n)100$

$n=5 \Rightarrow 2^7 \geq 600$ ×
$n=6 \Rightarrow 2^8 \geq 700$ ×
$n=7 \Rightarrow 2^9 \geq 800$ ×
$n=8 \Rightarrow 2^{10} \geq 900$ ✓

$n_{Least} = 8$ Ans

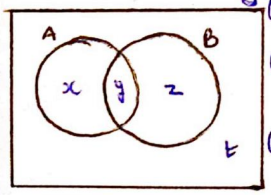
Information about Cards:

♣ ⇒ Club ♡ ⇒ Heart
♠ ⇒ Spade ◇ ⇒ Diamond
} 4 Suits of a deck of card

Honour Cards = 4 × 4 = 16 (A, J, Q, K)
Face Cards = 4 × 3 = 12 (J, Q, K)
Numbered Cards = 4 × 9 = 36 (2, 3, 4 - - 10)
Knave Cards = 4 × 3 = 12 (10, J, Q)

Set Theory:

2 Events :→



(#) $x+y+z+t = 1$

(#) Prob. of A and B = $P(A \cap B) = P(AB)$

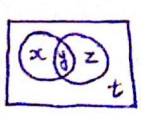
(#) Prob. of A or B = $P(A \cup B) = P(A+B)$
$= P(A) + P(B) - P(A \cap B)$

(#) If A & B are Mutually Exclusive,
$P(A \cap B) = y = 0$

(#) Prob. of non occurrence of A = $P(\bar{A}) = P(A^c)$
$= 1 - P(A)$

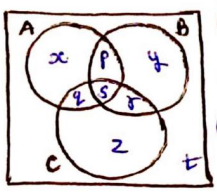
Ques Prob. that atleast one of the events A & B occurs is 0.6. If A and B occur together with prob. 0.2, then $P(A) + P(B) =$ ____.

Sol^n



$P(A \cup B) = x+y+z = 0.6$
$P(A \cap B) = y = 0.2$
$P(A \cup B) = P(A) + P(B) - P(A \cap B)$
$\Rightarrow P(A) + P(B) = 0.6 + 0.2 = 0.8$ Ans

3 Events :→



Exactly Two = p+q+r
Exactly One = x+y+z

(#) $x+y+z+p+q+r+s+t = 1$

(#) $P(A \cup B \cup C) = P(A)+P(B)+P(C) - P(A \cap B) - P(B \cap C) - P(A \cap C) + P(A \cap B \cap C)$

Eg $P(A \cap B \cap \bar{C}) = p$
(x, p, q, s) (x, p, y, t)
(y, p, s, r)

Ques For 3 events A, B & C

P(Exactly one of A or B occurs) = P(Exactly one of B or C occurs) = P(Exactly one of C or A occurs) $= \frac{1}{4}$, and P(All 3 occurs together) $= \frac{1}{16}$. Then prob. that atleast one of the events occurs = ____

Sol^n Jab 2 events ki baat kar rahe ho, toh teesre wale ko ignore karna hai while selecting region on Venn diagram.



$x+y+q+r = \frac{1}{4}$



$p+y+q+z = \frac{1}{4}$



$x+p+z+r = \frac{1}{4}$

Add all 3 eqns

$2(x+y+z+p+q+r) = \frac{3}{4} \Rightarrow$

Also,
$x+y+z+p+q+r+s+t = 1$
$\frac{6}{16} \quad \frac{3}{8} + \frac{1}{16} = 1-t = P(A \cup B \cup C)$

$\Rightarrow \boxed{\frac{1}{16} = P(A \cup B \cup C)}$ Ans

Independent Events:

2 Events are Independent when ⇒

$\boxed{P(A \cap B) = P(A) \cdot P(B)}$

→ If A & B are independent, then all possible combinations are also independent.

$(\bar{A}, B)$
$(A, \bar{B})$
$(\bar{A}, \bar{B})$
} All are Independent

→ Independent & mutually exclusive are two different things.

3 Events are Independent when ⇒

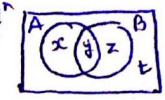
$P(A \cap B) = P(A) \cdot P(B)$
$P(B \cap C) = P(B) \cdot P(C)$
$P(A \cap C) = P(A) \cdot P(C)$
} Only pair wise independent when these 3 true

$P(A \cap B \cap C) = P(A) \cdot P(B) \cdot P(C)$

↳ Completely independent when all 4 are true simultaneously.

Ques If $P(\overline{A \cup B}) = 1/6$, $P(A \cap B) = 1/4$ and $P(\bar{A}) = \frac{1}{4}$, then comment on A & B.

Sol^n



$t = \frac{1}{6}$, $y = \frac{1}{4}$

$z+t = \frac{1}{4} \Rightarrow \boxed{z = \frac{1}{12}}$

$P(A) = x+y = \frac{9}{12}$; $P(B) = y+z = \frac{4}{12}$

$P(A \cap B) = \frac{3}{12} = P(A) \cdot P(B)$

$= \frac{9}{12} \times \frac{4}{12} \Rightarrow \frac{3}{12}$

$\therefore \boxed{P(A \cap B) = P(A) \cdot P(B)}$

↳ A and B are Independent and not Equally likely events.

Ques An exp. has 10 equally likely outcomes. Let A & B be two non-empty sets of the exp. If A consists of 4 outcomes, then how many outcomes B must have, so that A & B are independent.

Sol^n $S = \{ \text{---------} \}$ ← 10 outcomes

$A = \{ -, -, -, - \}$ $P(A) = 4/10 = \frac{2}{5}$

$B = \{ -, -, \text{----} \}$ Let, x outcome

For A & B to be Independent,

$P(A \cap B) = P(A) \cdot P(B) = \frac{2}{5} \cdot \frac{x}{10}$

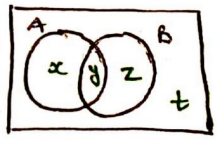
If $P(A \cap B) = 0 \Rightarrow$ Not possible
$P(A \cap B) = 1 \Rightarrow (x = 5/2)$ ×
$(A \cap B) = 2 \Rightarrow (x = 5)$ ✓
$(A \cap B) = 3 \Rightarrow (x = 15/2)$ ×
$(A \cap B) = 4 \Rightarrow (x = 10)$ ✓

$\boxed{x = 10, 5}$ Ans

Conditional Probability:

↳ Language: (Given that, Provided, Already occured)

Ek particular kaam ho chuka hoga, uske baad kisi dusre kaam ki probability puchi gayi hogi!



$P(A/B) = \frac{y}{y+z}$

$P(A/B)$ = 'B' ke hone ke baad 'A' ke hone ki prob.

= Ab B → Sample space hogya
↓
Usme se A ka fav. Region dekho.

Formula :→

$\boxed{P\left(\frac{A}{B}\right) = \frac{P(A \cap B)}{P(B)}}$ $\boxed{P\left(\frac{B}{A}\right) = \frac{P(B \cap A)}{P(A)}}$

Ques Assume that each born child is equally likely to be a boy or girl. If two families have two children each then the conditional prob. that all children are girls given that atleast 2 are girls is ____.

Probability - 2

Sol^n E_1 = Atleast 2 are girls
E_2 = All are girls



2 Girls → (BBGG) Prob. = $\left(\frac{1}{2}\right)^4 \times \frac{4!}{2!2!}$ (6)

3 Girls → (BGGG) Prob. = $\left(\frac{1}{2}\right)^4 \times \frac{4!}{3!} \Rightarrow \left(\frac{1}{2}\right)^4 \times 4$

All Girls → (GGGG) Prob. = $\left(\frac{1}{2}\right)^4$

$P(E_1) = 6\left(\frac{1}{2}\right)^4 + 4\left(\frac{1}{2}\right)^4 + \left(\frac{1}{2}\right)^4$

$= 11\left(\frac{1}{2}\right)^4$

$P(E_1 \cap E_2) = P(\text{All Girls}) = \left(\frac{1}{2}\right)^4$

So, $P\left(\frac{E_2}{E_1}\right) = \frac{P(E_1 \cap E_2)}{P(E_1)} = \frac{\left(\frac{1}{2}\right)^4}{11\left(\frac{1}{2}\right)^4}$

$\Rightarrow \boxed{P\left(\frac{E_2}{E_1}\right) = \frac{1}{11}}$ Ans

Imp

Ques If E, F & G are pairwise independent events with $P(E \cap F \cap G) = 0$ then $P\left(\frac{E \cap \bar{F}}{G}\right)$ is equal to ____.

Sol^n Pairwise Independent

↳ $P(E \cap F) = P(E) \cdot P(F)$
$P(F \cap G) = P(F) \cdot P(G)$
$P(E \cap G) = P(E) \cdot P(G)$
$P(E \cap F \cap G) = 0$

Now, $P\left(\frac{E \cap \bar{F}}{G}\right) = \frac{P(E \cap \bar{F} \cap G)}{P(G)}$ ← 'E' ke Bahar 'F' ke Bahar 'G' ke Andar



So, $P(E \cap \bar{F} \cap G) = P(G) - P(E \cap G) - P(F \cap G)$

$P(E \cap G) \Rightarrow \frac{P(G) - P(G)P(E) - P(G)P(F)}{P(G)}$

So, $P\left(\frac{E \cap \bar{F}}{G}\right) = 1 - P(E) - P(F)$

$= P(\bar{E}) - P(F)$ } Ans
$= P(\bar{F}) - P(E)$

Ques 1 Indian & 4 American couples are seated randomly on round table. Prob. that Indian man is seated adjacent to his wife, given that each American man is seated adjacent to his wife is ____.

Sol^n Let, E_1 → Indian couple sits together
E_2 → American couple sits together

↳ h, w (H1W1) (H2W2) (H3W3) (H4W4)

No. of ways = $(6-1)! \times 2^4$ ← Internal Arrangement

All couples together ⇒ $P(E_1 \cap E_2)$

(H1W1) (H1W1) (H2W2) (H3W3) (H4W4)

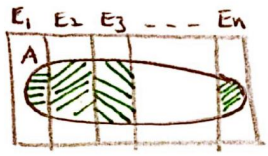
No. of ways = $(5-1)! \times 2^5$

So, $P\left(\frac{E_1}{E_2}\right) = \frac{4! \times 2^5}{5! \times 2^4} \Rightarrow \frac{2}{5}$ Ans

Total Probability & Baye's Theorem %

Conditional Probability

Suppose, there are n mutually exclusive events and an event A occurs through all of them in some proportion.



(Event A ho chuka hai, uski net prob. ↓ Given by T.P.T)

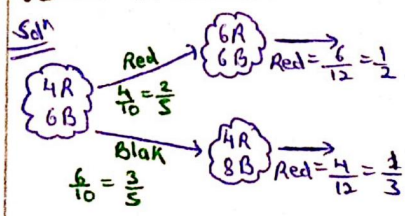
Total prob. of A = $P(E_1)P\left(\frac{A}{E_1}\right) + P(E_2)P\left(\frac{A}{E_2}\right) \cdots + P(E_n)P\left(\frac{A}{E_n}\right)$

(Even A ho chuka, batao uski E_3 ke through hone ki prob. kya hai?) → By B.T

Then, $P\left(\frac{E_3}{A}\right) = \frac{P(E_3 \cap A)}{P(A)}$

Ques A bag contains 4 Red & 6 Black balls. A ball is drawn, its colour is observed and this ball along with two more balls of same colour are returned to bag. If now, a ball is drawn, prob. that it will be Red is ____.

Sol^n



T.P.T

$P(\text{Red}) = \left(\frac{2}{5} \cdot \frac{1}{2}\right) + \left(\frac{3}{5} \cdot \frac{1}{3}\right) \Rightarrow \frac{2}{5}$ Ans

Ques Box 1 contains 30 cards, numbered 1-30 and Box 2 contains 20 cards numbered 31-50. A box is selected at random & non-prime card is drawn from it. Prob. that card is drawn from Box 1 is ____.

Sol^n → Box 1 Non Prime → $\frac{20}{30} = \frac{2}{3}$
(1, 2, .. 30)
↳ (2, 3, 5, 7, 11, 13, 17, 19, 23, 29)

→ Box 2 Non-Prime → $\frac{15}{20} = \frac{3}{4}$
(31, 32 .. 50) → (31, 37, 41, 43, 47)

By T.P.T

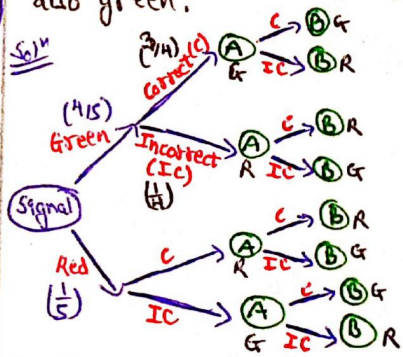
$P(\text{Non Prime}) = \left(\frac{1}{2} \cdot \frac{2}{3}\right) + \left(\frac{1}{2} \cdot \frac{3}{4}\right)$

By B.T

$P\left(\frac{\text{Box 1}}{\text{Non prime}}\right) = \frac{\frac{1}{2} \cdot \frac{2}{3}}{\frac{1}{2} \cdot \frac{2}{3} + \frac{1}{2} \cdot \frac{3}{4}} \Rightarrow \frac{8}{17}$ Ans

Ques A signal which can be Red or Green with prob. $\frac{1}{5}$ & $\frac{4}{5}$, is received by station A, then transmitted to station B. Prob. of each station receiving correct signal is $\frac{3}{4}$. If the signal received by station B is green, then find prob. that original signal was also green.

Sol^n



By T.P.T

Signal received is green = $\left(\frac{4}{5} \cdot \frac{3}{4} \cdot \frac{3}{4}\right) + \left(\frac{4}{5} \cdot \frac{1}{4} \cdot \frac{1}{4}\right) + \left(\frac{1}{5} \cdot \frac{3}{4} \cdot \frac{1}{4}\right) + \left(\frac{1}{5} \cdot \frac{1}{4} \cdot \frac{3}{4}\right)$

$= \frac{46}{80}$

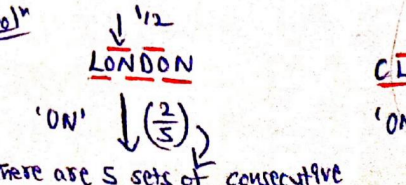
By B.T

$P\left(\frac{\text{Original was Green}}{\text{Green received at B}}\right) = \frac{\frac{4}{5} \cdot \frac{3}{4} \cdot \frac{3}{4} + \frac{4}{5} \cdot \frac{1}{4} \cdot \frac{1}{4}}{46/80}$

$= \frac{40/80}{46/80} \Rightarrow \frac{20}{23}$ Ans

Ques A letter come either from LONDON or CLIFTON. The postal mark on the letter legibly shows consecutive letters 'ON'. The prob. that letter has come from LONDON is ____.

Sol^n



(There are 5 sets of consecutive two letters, out of which 2 pairs has 'ON')

$P(\text{'ON' is visible}) = \left(\frac{1}{2} \cdot \frac{2}{5}\right) + \left(\frac{1}{2} \cdot \frac{1}{6}\right)$

$$P\left(\frac{\text{Letter came from London}}{\text{ON is visible}}\right) = \frac{\frac{1}{2}\cdot\frac{2}{5}}{\left(\frac{1}{2}\cdot\frac{2}{5}\right)+\left(\frac{1}{2}\cdot\frac{1}{6}\right)}$$

$$= \frac{12}{17} \text{ Ans}$$

Random Variable & Probability Distribution:

→ Random variable is denoted by 'X'

→ X is that whose Mean/Variance/Expected value/Probability Distribution is asked in the question.

Eg: If 3 coins are tossed simultaneously, then find prob. distribution of no. of heads.

Sol^n X = No. of Heads.

X_i's	$P(X_i)$
0	$(1/2)^3 = 1/8$
1	$(1/2)^3 \times 3 \Rightarrow 3/8$
2	$(1/2)^3 \times 3 \Rightarrow 3/8$
3	$(1/2)^3 \Rightarrow 1/8$

Sum of all prob. would be equal to 1 if prob. distribution is done correctly.

i.e. $\boxed{\sum P(X_i) = 1}$

Mean :→

→ Denoted as (μ) or $(\bar{X})$ = Expected value of X

$$\boxed{\mu = \bar{X} = E(X) = \sum P_i \cdot X_i}$$

Variance (σ_x^2) :→

→ Denoted as $V(X)$ or σ_X^2

$$\boxed{\sigma_x^2 = \sum P_i (X_i - \bar{X})^2 = \sum P_i X_i^2 - \left(\sum P_i X_i\right)^2} \quad (\bar{X})$$

Standard Deviation (σ_x): $\boxed{\sigma_x = \sqrt{\text{Variance}}}$

Ques Two cards are drawn ~~at random~~ successively with replacement from a well shuffled deck of 52 cards. Let X denote the random variable of no. of aces obtained in the two drawn cards. Then $P(X=1) + P(X=2)$ equals ——.

Sol^n X = No. of Aces (A's)

X_i	$P(X_i)$
0	$\left(\frac{48\times48}{52\times52}\right)$
1	$\left(\frac{4\times48}{52\times52}\right)$ Ace other + $\left(\frac{48\times4}{52\times52}\right)$ other Ace
2	$\left(\frac{4\times4}{52\times52}\right)$

So, $P(X=1) + P(X=2) = \frac{24}{169} + \frac{1}{169}$

$$= \frac{25}{169} \text{ Ans}$$

Ques A person throws two fair dice. He wins ₹15 for throwing a doublet, wins ₹12 ~~for~~ when the throw results in a sum of 9 and losses ₹6 for any other outcome. The expected gain/loss of the person is ——.

Sol^n X = Gain / Loss of Rs.

	D_1	D_2	
+15	1	1	$P = \frac{6}{36}$
	2	2	
	⋮	⋮	
	6	6	
+12	3	6	$P = \frac{4}{36}$
	6	3	
	5	4	
	4	5	
−6	Any other outcome		$P = \frac{26}{36}$

So, $M = 15\left(\frac{6}{36}\right) + 12\left(\frac{4}{36}\right) - 6\left(\frac{26}{36}\right)$

$$= \frac{1}{36}(90 + 48 - 156)$$

$$= \frac{1}{36} \times (-18) \Rightarrow -\frac{1}{2}$$

So, Loss of ₹ $\frac{1}{2}$ Ans